Tensor computations

# Generic identifiability of tensors at strictly subcritical ranks

**Difficulty:** extreme **Importance:** interesting to the community

**Status: Partially resolved**

**Rating rationale:** Extreme because the target is a classification across all formats and strictly subcritical ranks, with secant nondefectivity and exceptional families as obstacles; community importance is generic uniqueness of CP factors.

## Statement

Let $d \geq 3$, $n_1 \geq \cdots \geq n_d \geq 2$, and define

$$r_* = \frac{\prod_{i=1}^{d} n_i}{1 + \sum_{i=1}^{d}(n_i - 1)}, \qquad b = \prod_{i=2}^{d} n_i - \sum_{i=2}^{d}(n_i - 1).$$

For every integer $1 \leq r < r_*$, a generic rank-$r$ tensor in $\bigotimes_{i=1}^{d} \mathbb{C}^{n_i}$ is conjectured to have a unique expression as a sum of $r$ nonzero rank-one tensors, except in these cases:

1. $n_1 > b$ and $r \geq b$.
2. Format $(4, 4, 3)$, rank 5.
3. Format $(n, n, 2, 2)$, rank $2n - 1$, with $n \geq 2$.
4. Format $(4, 4, 4)$, rank 6.
5. Format $(6, 6, 3)$, rank 8.
6. Format $(2, 2, 2, 2, 2)$, rank 5.

A rank-one summand is $v_1 \otimes \cdots \otimes v_d$. Uniqueness identifies decompositions whose tensor summands differ only by permutation. Generic means outside a proper Zariski-closed subset of the Zariski closure of tensors of rank at most $r$, restricted to tensors of actual rank $r$. The number $r_*$ is a rational dimension-count value, not an assumed actual generic rank.

## Relevance

The conjecture specifies the range in which a generic low-rank tensor model has uniquely recoverable factors, an essential precondition for numerical CP decomposition and component interpretation.

### References

1. L. Chiantini, G. Ottaviani, and N. Vannieuwenhoven, *Effective criteria for specific identifiability of tensors and forms*. Primary preprint, §3, Conjecture 6 and Remark 7, p.7. The author journal manuscript numbers these Conjecture 3.4 and Remark 3.5.
2. The same authors, *An Algorithm for Generic and Low-Rank Specific Identifiability of Complex Tensors*, SIAM J. Matrix Anal. Appl. 35 (2014), 1265–1287. DOI; primary preprint, Theorem 1.1, for the verified bounded-format classification.
3. A. Massarenti and M. Mella, *Bronowski's conjecture and the identifiability of projective varieties*. January 2024 primary revision, abstract and §§3–4, for later conditional links between identifiability and nondefectivity.

### Status check — 2026-09-10

Rechecked Chiantini–Ottaviani–Vannieuwenhoven, Conjecture 6 and Remark 7, including all six exceptions. The source records proof for formats with product of dimensions at most 15000. Targeted searches for later Segre identifiability and classification results found additional nondefectivity/identifiability bounds, but no unrestricted classification. The displayed family therefore has substantive proved cases and an unresolved general remainder; a fresh primary reaffirmation of the entire list was not located.